

RetailPulse Executive Report

Sales Snapshot

date	product_id	quantity_sold	revenue	orders	product_name	category
2025-12-26	P001	5	63.0	3	Everyday Coffee	Grocery
2025-12-26	P002	10	62.95	5	Organic Cereal	Grocery
2025-12-26	P003	7	124.38	3	Cotton T-Shirt	Apparel
2025-12-26	P004	2	151.98	2	Running Shoes	Footwear
2025-12-26	P005	18	594.82	10	Desk Lamp	Home
2025-12-26	P006	12	288.63	4	Wireless Mouse	Electronics
2025-12-26	P007	2	56.98	2	Yoga Mat	Fitness
2025-12-26	P008	7	98.93	4	Water Bottle	Fitness
2025-12-27	P001	9	111.71	4	Everyday Coffee	Grocery
2025-12-27	P002	8	51.27	4	Organic Cereal	Grocery
2025-12-27	P003	14	255.41	4	Cotton T-Shirt	Apparel
2025-12-27	P004	8	639.92	2	Running Shoes	Footwear

Forecast

date	yhat	lstm_yhat	ensemble_yhat	yhat_lower	yhat_upper
2026-06-24	48.07	59.39	52.03	40.85759143370918	55.27791782207713
2026-06-25	48.72	59.41	52.46	41.412090874143196	56.02812294737021
2026-06-26	49.37	59.47	52.9	41.96676273199898	56.77856134329274
2026-06-27	50.03	59.51	53.35	42.52166169902262	57.52930700456001
2026-06-28	50.68	59.54	53.78	43.0768423996628	58.28043383483791
2026-06-29	45.26	59.55	50.26	38.468978365292784	52.04626484716082
2026-06-30	45.91	59.57	50.69	39.02488605641555	52.798375252797506
2026-07-01	47.82	59.58	51.94	40.64684453432067	54.9927896640809
2026-07-02	48.47	59.58	52.36	41.20369650199617	55.74617762034775
2026-07-03	49.13	59.59	52.79	41.76110175051381	56.50031413304809
2026-07-04	49.79	59.59	53.22	42.319114161657	57.25527210106535
2026-07-05	50.44	59.59	53.64	42.877787437293776	58.01112417986804

Inventory Recommendations

time_days	stock_on_hand	holding_cost	ordering_cost	service_level	avg_daily_demand	demand_std	annual_demand	safety_stock	reorder_point	eoq
	135	2.35	59	0.98	7.79	4.26	2844.48	78.15	701.6	377.93
	351	1.34	75	0.9	6.78	4.45	2473.19	62.4	875.5	526.16
	120	3.42	82	0.9	7.15	4.32	2609.23	42.79	471.7	353.72
	338	12.9	57	0.95	7.44	4.41	2715.35	39.9	263.08	154.91
	205	5.51	59	0.9	7.5	4.65	2737.5	39.9	377.4	242.13
	113	4.06	56	0.98	6.94	4.25	2531.66	61.56	408.37	264.27
	153	5.55	124	0.95	6.64	4.11	2424.19	40.09	272.55	329.13
	232	3.1	102	0.9	6.99	4.08	2550.73	49.54	678.49	409.7